

Sieng Lyheng

Applied Mathematics & Statistics | Data Science Student

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Education

Institute of Technology of Cambodia

Phnom Penh, Cambodia

B.Eng. Applied Mathematics & Statistics (Data Science)

Expected 2027

Year 3 GPA: **4.00 / 4.00**

Experience

Data Visualization for Library Resources & Attendance

Aug 2024 – Sep 2024

ITC Library, Informal Internship

Phnom Penh, Cambodia

- Analyzed book-borrowing and room-booking data from multiple departments using Python and Jupyter Notebook.
- Reported findings and insights to the Head of Library to support data-driven decision-making.

Projects

Data Engineering

Real-Time Weather API & Analytics System

2026

Python, Kafka, PySpark, PostgreSQL, Docker, Streamlit

- Built a real-time pipeline that collects WeatherAPI data through Kafka and processes it with PySpark Structured Streaming.
- Designed Bronze, Silver, and Gold layers in PostgreSQL and developed dashboards for pipeline monitoring and weather analytics.

Cybersecurity Attacks ETL & Analytics Pipeline

2026

Python, Pandas, SQL Server, PySpark, JDBC, Power BI

- Developed a batch ETL pipeline to clean and load approximately 40,000 cybersecurity records into Microsoft SQL Server.
- Used PySpark through JDBC for analysis and built a Power BI dashboard for attack trends, severity, protocols, and actions.

Data Analytics & Machine Learning

Loan Default Risk Analysis and Prediction

2026

Python, Pandas, Scikit-learn, Random Forest, Streamlit, Power BI

- Built a credit-risk workflow to analyze applicant data and predict loan-default likelihood.
- Applied feature engineering and Random Forest classification, then delivered a Streamlit application and Power BI dashboard.

Customer Retention & Churn Analysis Dashboard

2026

Power BI, Power Query, DAX, Excel

- Built a three-page dashboard to analyze churn, retention, customer profiles, and behavior across 5,630 customers.
- Cleaned data with Power Query and created DAX measures for churn, retention, revenue, and customer KPIs.

Projects Continued

Artificial Intelligence & Computer Vision

Sao Somleng Game — Khmer Voice Learning Game2026

Python, FastAPI, PyTorch, Wav2Vec2, React

- Developed a voice-controlled educational game that helps children practice Khmer character pronunciation.
- Fine-tuned Wav2Vec2 speech models and integrated a continuous-play mode with performance tracking.

Automatic License Plate Recognition in Cambodia (ALPRC)2025

Python, YOLOv8, PaddleOCR, EasyOCR, Roboflow, Flask

- Developed an AI system for detecting and recognizing Cambodian vehicle license plates.
- Used YOLOv8 for plate detection and compared PaddleOCR and EasyOCR for text-recognition performance.

Project Exhibitions

Automatic License Plate Recognition in Cambodia (ALPRC)

- Demonstrated and exhibited the system to visitors at the 14th Scientific Day 2025, The Use of Artificial Intelligence in Education 2025, and the Scientific Day 2026.
- Explained the project workflow, system architecture, and practical applications, and answered technical questions from visitors.

Technical Skills

ProgrammingPython, R, SQL, Java, C, C++

Data EngineeringKafka, PySpark, Structured Streaming, ETL, Medallion Architecture, Docker, SQLAlchemy, JDBC

Data AnalyticsPandas, Power Query, DAX, Data Cleaning, Exploratory Data Analysis, KPI Development

Machine LearningScikit-learn, TensorFlow/Keras, PyTorch, Random Forest, Feature Engineering, Classification, Deep Learning

Computer VisionYOLOv8, OpenCV, PaddleOCR, EasyOCR, Roboflow, CNN

VisualizationPower BI, Microsoft Excel, Matplotlib, Streamlit, Jupyter Notebook

DatabasesMicrosoft SQL Server, PostgreSQL, MySQL, MongoDB

ToolsGit, GitHub, VS Code, Docker Compose, pgAdmin, SQL Server Management Studio, MySQL Workbench

Web & APIsFastAPI, Flask, REST APIs, React

Languages

Khmer (Native) · English (B2)